Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program
(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Teaching Robots to Weld through Natural Language Dialogue and Demonstration*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Mechanical Materials		Mechanical Materials	
	Medium Category	**Small Category**	**Medium Category**	**Small Category**
	Robotics Automation Machinery	Robot Control and Intelligence Technologies	Casting/Welding	Casting/Welding-related Software
Concept of Proposed Technology	A cognitive robotics system that enables operators, integrators, and end users to quickly and efficiently reprogram industrial robots through natural language dialogue and physical demonstration, eliminating long delays and the need for specialized programming expertise. The system provides transparent and explainable decision-making using proprietary OntoAgentic AI, allowing users to verify robot understanding and trace reasoning steps. Unlike generative AI approaches that produce plausible-sounding outputs but lack genuine understanding and reliability, HARMONIC delivers content-based semantic interpretation that meets the safety and trust requirements of manufacturing environments. The			

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	technology targets welding applications in high-mix, low-volume manufacturing, where frequent task changes make traditional programming economically infeasible for SMEs.
Rationale for Joint R&D Project	Through a prior customer discovery process in the framework of the NSF National I-Corps Program, we validated strong market demand for HARMONIC's approach to robot reprogramming. However, customer discovery also revealed the key challenge for commercialization: deploying HARMONIC to a new application domain requires extending the knowledge resources, like ontology, lexicon, and procedural knowledge, to cover domain-specific terminology, processes, and task sequences. We have developed and tested the acquisition process. This knowledge acquisition cannot be done in isolation. Collaboration with an industry partner is essential because domain-specific knowledge involving specific vocabulary, process parameters, quality criteria, and operator workflows can only come from practitioners with real manufacturing experience. We chose welding as the target application because it represents one of the most complex operations a robot can perform in manufacturing, involving numerous interrelated variables, strict quality requirements, and skilled judgment. If HARMONIC can succeed in welding, it can be extended to other manufacturing applications with confidence. The welding industry also faces a severe skilled labor shortage, making automation critical. Yet the programming bottleneck means skilled welders spend time struggling with complex programming interfaces rather than applying their welding expertise directly to a cognitive robot capable of learning from instruction and demonstration. Furthermore, welding job shops and custom fabricators represent exactly the high-mix, low-volume environments where traditional robot (re)programming is economically unfeasible, and where HARMONIC's value proposition is strongest. A Korean SME partner provides the application context, use-case definitions, sample parts, and subject-matter expertise necessary to efficiently extend HARMONIC's knowledge base and to validate the system in real manufacturing conditions, a collaboration that neither party could achieve independently.
Key Words	1. Cognitive Robotics 2. Explainable AI 3. Welding Automation 4. Natural Language Robot Programming 5. OntoAgentic AI

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)

R&D Objectives

1. **Welding Process Understanding and Adaptation:** Extending HARMONIC's knowledge resources to cover the welding domain. This involves:
   - Expanding HARMONIC's existing ontological world model to include welding-specific concepts: weld types (fillet, butt, tack), joint geometries, process parameters (voltage, wire feed speed, travel speed), and quality criteria.
   - Developing domain-related semantic lexicon entries that map welding vocabulary to ontological concepts.
   - Creating procedural knowledge (ontological scripts), capturing welding task sequences, and operator workflows.
   - Implementing quality feedback integration where operators can flag defects and suboptimal operation verbally, allowing the system to learn during execution through parameter adjustments.
2. **Integration of Cognition and Control for Welding:** Adapting and enhancing the existing bidirectional interface between HARMONIC's strategic (cognitive) layer and the industry partner's robot control systems. The partner provides the robot, controllers, and domain expertise, while RPI adapts the integration for welding-specific requirements. This involves:
   - Translating abstract action commands from OntoAgent (e.g., "weld a fillet along this joint") into precise, executable robot control operations.
   - Quantifying qualitative verbal instructions (e.g., "it's running a bit cold," "reduce spatter") into adjustable control parameters.
   - Adapting flexible scheduling to manage real-time reactive responses required in welding while maintaining deliberative planning capabilities.
3. **Multilingual Support:** Expanding HARMONIC's language capabilities to support Korean in addition to English. HARMONIC's semantic lexicon architecture currently contains approximately 29,000 English entries that map vocabulary to ontological concepts. Since the underlying ontological concepts remain constant across languages, only the lexical entries need to be acquired for each new language. This allows for systematic extension to Korean, enabling operators to converse naturally in their native language. Lexical entries will be acquired not only for welding processes but also for specific human operator requirements and conventions.
4. **Validation in Real Welding Scenarios**: Testing and validating the integrated system in real manufacturing conditions, including operator feedback loops, iterative refinement, and demonstration of reprogramming tasks across multiple weld types and part geometries.

R&D Target Specifications

Specification	Target
Knowledge acquisition	Ontological world model expansion to cover welding-specific concepts, including weld types (fillet, butt, tack, lap), joint geometries, process parameters (voltage, wire feed speed, travel speed, torch angle), quality criteria, and defect types. Development of semantic lexicon entries mapping welding vocabulary to ontological concepts for both English and Korean.
Long-horizon	Robot acquires multi-step welding task sequences (more than 10 steps) with varying levels of complexity through natural language

task learning	instruction and demonstration. Operators teach complex procedures involving multiple weld segments, repositioning, parameter changes, and quality checks through conversational dialogue.
Verbal instruction types	Support for task specification (e.g., "weld a fillet along this joint"), parameter adjustment (e.g., "it's running a bit cold," "reduce spatter on this pass"), and quality feedback (e.g., "there's porosity here, adjust the settings"). The system learns from operator corrections and adapts parameters accordingly.
Robot platform integration	Successful integration with the collaborator's robot platform, adapting HARMONIC's cognitive-control interface to the specific controllers, motion planners, and welding equipment provided by the industry partner.
Multilingual support	English and Korean language capabilities, leveraging HARMONIC's OntoAgent architecture.
Demonstration and validation	Demonstration of robot reprogramming through natural language and demonstration in real-world welding scenarios, including operator testing, iterative refinement based on feedback, and validation across multiple part geometries and weld types.

R&D Plan

The proposed project spans 36 months, organized into three phases of approximately one year each. The plan progressively builds from system integration and foundational work, through knowledge acquisition and proof-of-concept development, to multilingual expansion and real-world validation. This phased approach ensures systematic development while allowing for iterative refinement based on feedback from the industry partner and operator testing.

Phase 1 (Year 1): Integration and Foundation

This phase focuses on establishing the foundation for the cognitive-robotic welding system. Key activities include:

- Specification procurement and requirements gathering in collaboration with the industry partner, including identification of target welding applications, robot platform specifications, and operator workflow requirements.
- Integration of HARMONIC's strategic (cognitive) layer with the collaborator's robot control systems, adapting the bidirectional interface APIs for the specific controllers, motion planners, and welding equipment.
- Research and development of integration strategies for translating abstract cognitive commands into precise robot control operations, including methods for quantifying qualitative verbal instructions into control parameters.
- Initial knowledge acquisition planning, identifying the welding-specific concepts, terminology, and procedural knowledge required for the target applications.

Phase 2 (Year 2): Knowledge Acquisition and Proof-of-Concept Development

This phase focuses on building the welding domain knowledge and demonstrating core capabilities. Key activities include:

- Systematic knowledge acquisition for the welding domain, expanding the ontology to cover weld types, joint geometries, process parameters, quality criteria, and defect types.
- Development of semantic lexicon entries mapping welding vocabulary to ontological concepts.
- Creation of models for acquiring procedural knowledge (scripts), capturing welding task sequences and operator workflows.
- Development of proof-of-concept systems demonstrating long-horizon task learning through natural language instruction and demonstration.
- Implementation and testing of verbal instruction capabilities, including task specification, parameter adjustment, and quality feedback.
- Iterative refinement based on operator feedback using the robot platform provided by the collaborator.

Phase 3 (Year 3): Multilingual Expansion and Validation

This phase focuses on expanding language capabilities and validating the system in real-world conditions. Key activities include:

- Expansion of HARMONIC's lexicon to support Korean language.
- Acquisition of Korean lexical entries specific to welding processes and operator conventions.
- Comprehensive validation testing in real-world welding scenarios across multiple part geometries and weld types.
- Operator testing with both English and Korean-speaking users.
- System refinement and improvements based on validation results.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Language, Embodiment, Intelligence, and Agency (LEIA) Lab at Rensselaer Polytechnic Institute will serve as the primary research and development partner, contributing expertise in cognitive robotics, computational semantics, and human-robot interaction developed over three decades of research. The LEIA Lab will contribute the HARMONIC cognitive-robotic architecture, including the OntoAgent cognitive system, the OntoAgentic AI infrastructure, the bidirectional interface APIs connecting strategic and tactical layers, and the development environment (DEKADE) for knowledge acquisition and system testing. The lab brings an established foundation of knowledge resources, an ontological world model of approximately 9,000 concepts, an English semantic lexicon of approximately 29,000 entries, and proven processing modules for natural language understanding, reasoning, and decision-making. The LEIA Lab will lead research activities across all three project phases: developing integration strategies for connecting HARMONIC with the collaborator's robot control systems; extending the ontology, lexicon, and procedural knowledge to cover the welding domain; expanding language capabilities to support Korean; and conducting iterative refinement based on validation testing. The lab will also lead the development of methods for quantifying qualitative verbal instructions into control parameters—a core research challenge for cognitive-control integration. The LEIA Lab will provide access to HARMONIC interfaces and APIs as needed for integration with the collaborator's robot platform and the DEKADE development and demonstration environment and will bring in expertise in knowledge acquisition methodology. The lab will also engage personnel, including faculty researchers with expertise in cognitive architectures, computational semantics, and robotic systems.
Expected R&R of Korean SME Partner	The Korean industry partner will provide the robotic infrastructure essential for this collaboration, including collaborative robot hardware platforms, control software, and welding equipment such as welding torches, wire feeders, and power sources. They will contribute domain expertise from their welding application experience, providing the specific vocabulary, process parameters (voltage, wire feed speed, travel speed, torch angles), joint geometries, quality criteria, and operator workflows necessary for HARMONIC's knowledge acquisition. The partner will also share technical documentation related to their robot control architecture to support integration efforts. The industry partner will participate in joint development activities, particularly in adapting HARMONIC's cognitive-control interface to their specific robot controllers and motion planners. They will collaborate in developing demonstrations that showcase robot reprogramming through natural language and demonstration in realistic welding scenarios. The partner will provide access to facilities, equipment, sample parts, and workpieces representing target welding applications. They will lead testing and validation efforts, coordinating operator testing with both English and Korean speaking welders in real manufacturing scenarios, collecting feedback on system usability and performance, and providing iterative input for system refinement throughout the project phases.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Achievements:** This joint R&D project will produce a validated cognitive-robotic system capable of learning welding tasks through natural language instruction and demonstration. Key technological outcomes include: an expanded ontology covering welding-specific concepts, processes, and quality criteria; a bilingual (English and Korean) semantic lexicon enabling operators to converse naturally in their native language; proven integration of HARMONIC's cognitive architecture with collaborative robot platforms for welding applications; and demonstrated long-horizon task learning (more than 10 steps) with varying levels of complexity. The project will advance the state of the art in cognitive-control integration, particularly in methods for quantifying qualitative verbal instructions into precise robot control parameters. **Commercialization Potential:** Prior customer discovery validated strong market demand for this technology. Integrators spend 30–50% of project time on customer-specific robot programming, and reprogramming costs can exceed $100,000 per deployment. The validated value proposition of reducing robot reprogramming costs by enabling operators to reliably teach robots through dialogue addresses a significant industry pain point. The technology targets industrial automation integrators seeking to reduce deployment time, robot original equipment manufacturers seeking competitive differentiation, and manufacturing end users, particularly small and medium enterprises, who currently cannot justify robotic automation due to programming overhead. **Industrial and Economic Impact:** The welding industry faces a severe skilled labor shortage while demand for welding continues to grow. This technology enables skilled welders to focus on applying their welding expertise rather than struggling with complex programming interfaces. By reducing reprogramming time from days to hours and lowering minimum viable batch sizes, the technology makes robotic welding economically viable for high-mix, low-volume manufacturing environments that currently cannot adopt automation. This expands access to robotic automation for small and medium enterprises, driving productivity gains and competitiveness across the manufacturing sector in both South Korea and the United States.

6. References

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(2) McShane, Marjorie, and Sergei Nirenburg. *Linguistics for the Age of AI*. Mit Press, 2021.

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□ **Terms:**

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization